

ACT / ALOHA · arXiv 2304.13705 · RSS 2023

Archaeologist

11+3 · weight 5

Priors $\rightarrow$ ACT $\parallel$ DP $\rightarrow$ π_0 $\rightarrow$ RTC
 $\rightarrow$ BEHAVIOR · densest deck

Oct 12 paper club · VLA Architectures

Spine: action chunking as motor contract — TE $\neq$ receding $\neq$ open-loop $\neq$ soft-inpaint

Zhao, Kumar, Levine, Finn · Stanford / Berkeley / Meta

Talk arc · Archaeologist

1 Priors

BC · DAgger
chunk psych
CVAE · TF

2 Pivot

ACT+ALOHA
chunk+TE on
cheap HW

3 Twin '23

|| Diffusion
Policy
receding

4 Downstream

Mobile/ALOHA2
 $\pi_0/\pi_{0.5}$
RTC

5 BEHAVIOR

30/26/4
corr. FM
'26 baselines

Honesty aloud

“Chunking is the shared prior. Execution dialects changed. CVAE is optional nostalgia.”

Priors → ACT/ALOHA · CLEAR inputs

Novelty = popularizing chunking+TE on open low-cost bimanual — not inventing transformers/CVAE

Prior	Relation	Conf.
BC / ALVINN	Supervised obs→action IL framing	CLEAR
Dagger compounding	Motivates chunking vs on-policy fix	CLEAR
Lai et al. 2022 psych	Action chunking as policy compression	CLEAR
Transformers / BERT	Sequence backbone + [CLS] encoder	CLEAR
CVAE / β -VAE	Generative action-sequence model	CLEAR
BeT · RT-1 · VINN	Direct baselines (often no chunks)	CLEAR
Joint-space teleop	Leader-follower hardware ancestry	CLEAR
ResNet-18 towers	Multi-cam visual encoders	CLEAR

This paper's pivot

Popularized action chunking + temporal ensembling

for high-Hz visuomotor BC on open low-cost bimanual hardware

ACT = CVAE + Transformer implementation with strong fine-manipulation evidence

Hardware

ALOHA <\$20k
50 Hz joint teleop
4 RGB · PID joints

Algorithm

Predict $k \times 14$ chunks
TE every-step ensemble
CVAE for human noise

Evidence

Open Cup 84%
Battery 96% · Shoe 92%
~10-20 min demos

Not a foundation model — per-task ~80M from scratch

Fig 4 — ACT CVAE encoder + Transformer policy

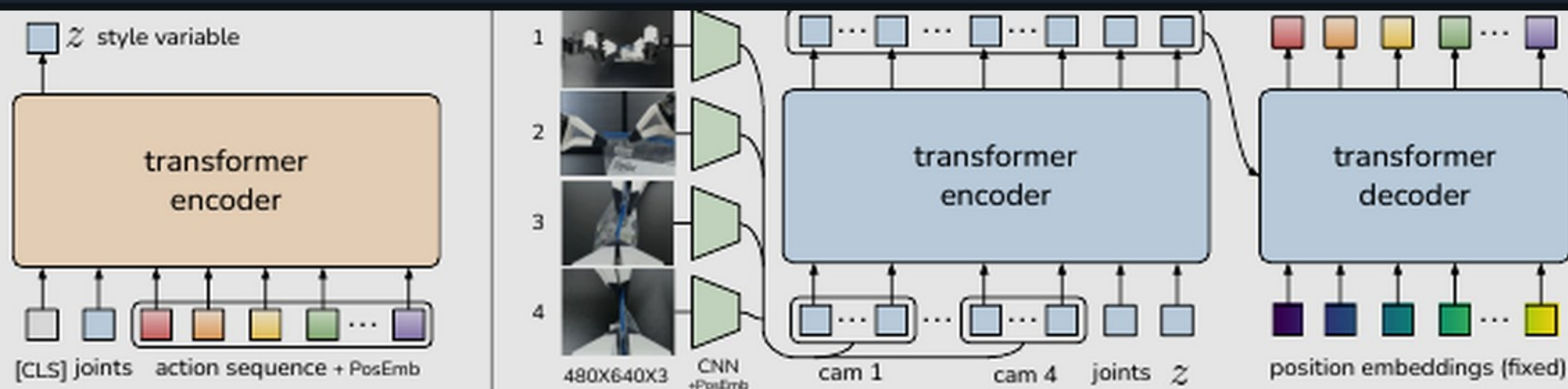


Fig. 4: *Architecture of Action Chunking with Transformers (ACT)*. We train ACT as a Conditional VAE (CVAE), which has an encoder and a decoder. *Left*: The encoder of the CVAE compresses action sequence and joint observation into z , the style variable. The encoder is discarded at test time. *Right*: The decoder or policy of ACT synthesizes images from multiple viewpoints, joint positions, and z with a transformer encoder, and predicts a sequence of actions with a transformer decoder. z is simply set to the mean of the prior (i.e. zero) at test time.



Fig. 5: We employ both Action Chunking and Temporal Ensembling when applying actions, instead of interleaving observing and executing.

To train ACT on a new task, we first collect human demonstrations using *ALOHA*. We record the joint positions of the leader robots (i.e. input from the human operator) and use them as actions. It is important to use the leader joint positions instead of the follower's, because the amount of force applied is implicitly defined by the difference between them, through the low-level PID controller. The observations are composed of the current joint positions of follower robots and the image feed from 4 cameras. Next, we train ACT to predict the *sequence of future actions* given the current observations. An action here corresponds to the target joint positions for both arms in the next time step. Intuitively, ACT tries to imitate what a human operator would do in the following time steps given current

rable to a single research arm such as Franka Emika Panda. *ALOHA* enables the teleoperation of:

- **Precise tasks** such as threading zip cable ties, picking credit cards out of wallets, and opening or closing ziploc bags.
- **Contact-rich tasks** such as inserting 288-pin RAM into a computer motherboard, turning pages of a book, and assembling the chains and belts in the NIST board #2 [4]
- **Dynamic tasks** such as juggling a ping pong ball with a real ping pong paddle, balancing the ball without it falling off, and swinging open plastic bags in the air.

Skills such as threading a zip tie, inserting RAM, and juggling ping pong ball, to our knowledge, are not available for existing teleoperation systems with 5-10x the budget [21] [5]. We include a more detailed price & capability comparison in Appendix A, as well as more skills that *ALOHA* is capable of in Figure 9. To make *ALOHA* more accessible, we open-source all software and hardware with a detailed tutorial covering 3D printing, assembling the frame to software installations. You can find the tutorial on the [project website](#).

operator would do in the following time steps given current observations. These target joint positions are then tracked by the low-level, high-frequency PID controller inside Dynamixel motors. At test time, we load the policy that achieves the lowest validation loss and roll it out in the environment. The main challenge that arises is compounding errors, where errors from previous actions lead to states that are outside of training distribution.

A. Action Chunking and Temporal Ensemble

To combat the compounding errors of imitation learning in a way that is compatible with pixel-to-action policies (Figure II), we seek to reduce the effective horizon of long trajectories collected at high frequency. We are inspired by *action chunking*, a neuroscience concept where individual actions are grouped together and executed as one unit, making them more efficient to store and execute [35]. Intuitively, a chunk of actions could correspond to grasping a corner of the candy wrapper or inserting a battery into the slot. In our implementation, we fix the chunk size to be k : every k steps, the agent receives

Action chunking vs Temporal Ensembling

Naïve open-loop every k steps vs every-step query + exp-weighted overlap

Naïve chunking (open-loop)

observe → predict k → execute all k → replan · jerky at boundaries



ACT Temporal Ensembling

Query every tick · buffer overlapping preds for same absolute t · $a_t = \sum w_i \hat{A}_t[i] / \sum w_i$ · $w_i = \exp(-m \cdot i)$



Same-t fusion

$$w = [e^0, e^{-m}, e^{-2m}, \dots]$$

recent obs → faster (small m)

+~3.3 pp ACT (paper)

Control schemes — do NOT conflate

Chunking is shared · TE ≠ receding ≠ open-loop ≠ soft-inpaint

Scheme	Mechanism	Commit / loop
ACT TE	Every-step query; exp-avg overlapping preds for same t	ensembled 1-step @ 50 Hz
DP receding	Predict T_p , exec $T_a < T_p$, replan	$T_a \approx 8$ @ ~ 10 Hz
π_0 open-loop	$H=50$; exec prefix; NO TE (hurt early for them)	open-loop prefix @ 20-50 Hz
RTC soft-inpaint	Async; freeze prefix; soft-mask remainder	latency-aware async
BEHAVIOR'25	Predict 30 / exec 26 / keep 4 + correlated FM	receding ~ 30 Hz

Map: $k/50 \leftrightarrow DP\ T_a \cdot \Delta t \leftrightarrow \pi_0/Comet\ H/Hz$ · Defaults: $k=100$ ($\sim 2s$ @ 50 Hz); $L1+\beta KL$; test $z=0$

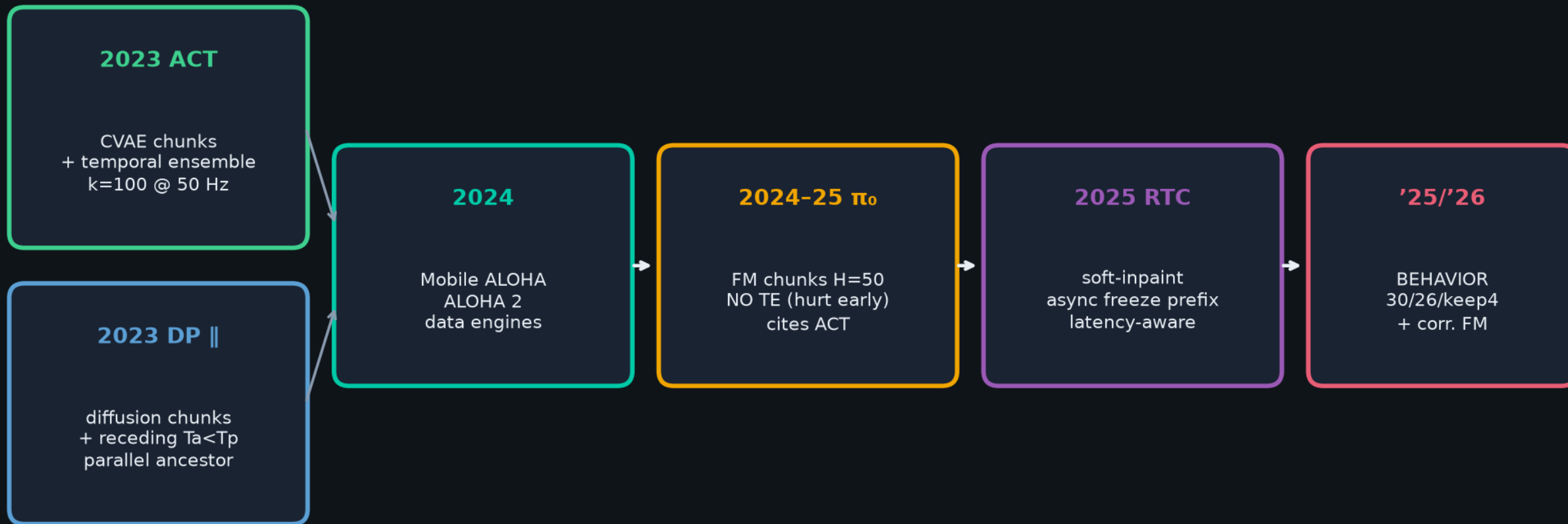
Horizon map · normalize by commit seconds

ACT	$k=100$ @ 50 Hz	~2.0 s chunk	TE → 1-step ensembled
DP	$T_p=16, T_a=8$ @ 10 Hz	~0.8 s commit	receding replan
π_o	$H=50$ @ 20-50 Hz	~1.0-2.5 s	open-loop prefix
BEHAVIOR'25	30 / exec 26 @ ~30 Hz	~0.87 s	keep 4 + inpaint

$k/50 \leftrightarrow T_a \cdot \Delta t \leftrightarrow H/\text{Hz}$ — Empiricist + Archaeologist shared language

Genealogy · CLEAR lineage

ALOHA+ACT (CVAE+TE) || DP → Mobile/ALOHA2 → $\pi_0/\pi_{0.5}$ → RTC → BEHAVIOR'25/'26



STAYED: generative continuous chunks · overlap / consistency at boundaries

MOVED: CVAE→diffusion→flow · TE→receding→soft-inpaint · tabletop→mobile→household VLA

Club line

ACT coined chunk + ensemble for 50 Hz fine bimanual BC

Later VLAs keep chunks, often drop ensemble,

retune H / commit Hz · TE ≠ receding ≠ open-loop ≠ soft-inpaint

ACT+TE



DP recede



π_0 open



RTC inpaint

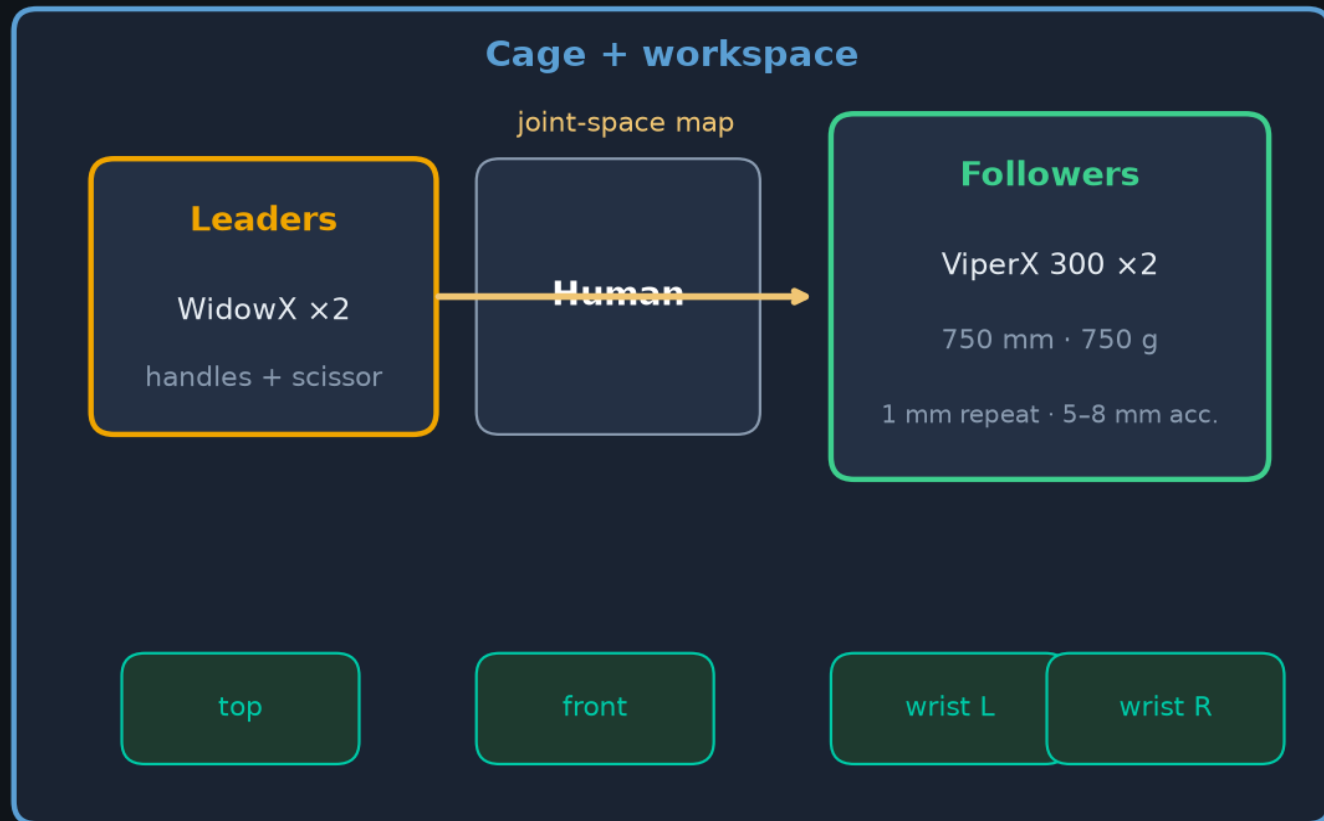


BEH 30/26/4

Map horizons by commit_seconds — not raw k or H

ALOHA hardware · <\$20k leader-follower bimanual

WidowX leaders → ViperX followers · joint-space teleop · 4 RGB · 50 Hz



Budget

<\$20k total
2xViperX + 2xWidowX
+ cage + cams

Control

50 Hz record/control
Absolute joint targets
Dynamixel PID

Design

Low-cost · versatile
Repairable · <2 h build
Rubber-band gravity assist

Follow-ons: Mobile ALOHA (2401.02117) · ALOHA 2 (2405.02292) — data engines, not just demos

Genealogy = control habit, not CVAE nostalgia

ALOHA+ACT (CVAE+TE) || DP → Mobile/ALOHA2 → $\pi_0/\pi_{0.5}$ → RTC → BEHAVIOR'25/'26

Chunking shared · TE ≠ receding ≠ open-loop ≠ soft-inpaint

Q&A · keep the spine clear

Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware

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¹ Stanford University ² UC Berkeley ³ Meta

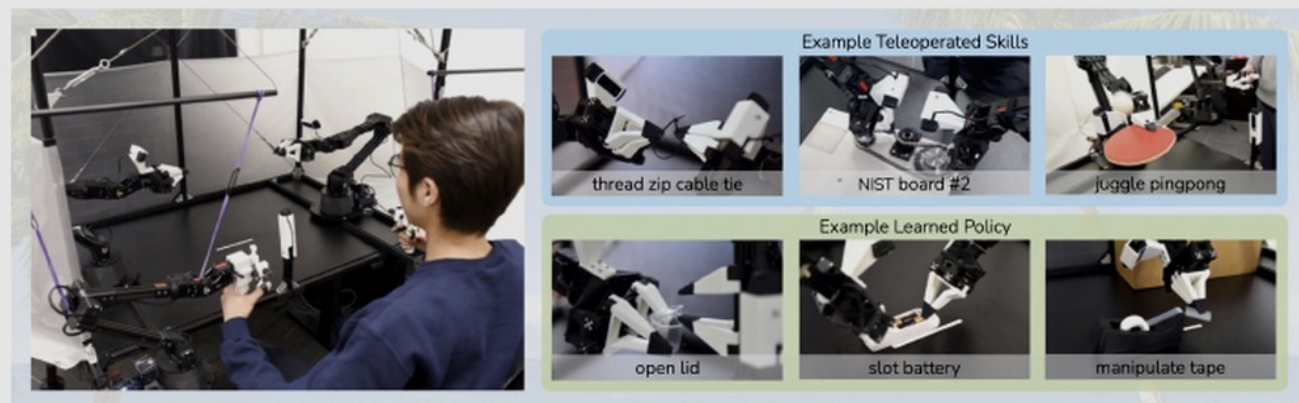


Fig. 1: ALOHA 🌈 : A Low-cost Open-source Hardware System for Bimanual Teleoperation. The whole system costs <\$20k with off-the-shelf robots and 3D printed components. Left: The user teleoperates by backdriving the leader robots, with the follower robots mirroring the motion. Right: ALOHA is capable of precise, contact-rich, and dynamic tasks. We show examples of both teleoperated and learned skills.

Abstract—Fine manipulation tasks, such as threading cable ties or slotting a battery, are notoriously difficult for robots because they require precision, careful coordination of contact forces, and closed-loop visual feedback. Performing these tasks typically requires high-end robots, accurate sensors, or careful calibration, which can be expensive and difficult to set up. *Can learning enable low-cost and imprecise hardware to perform these fine manipulation tasks?* We present a low-cost system that performs end-to-end imitation learning directly from real demonstrations, collected with a custom teleoperation interface. Imitation learning, however, presents its own challenges, particularly in high-precision domains: errors in the policy can compound over time, and human demonstrations can be non-stationary. To address these challenges, we develop a simple yet novel algorithm, *Action Chunking with Transformers (ACT)*, which learns a generative model over action sequences. ACT allows the robot to learn 6 difficult tasks in the real world, such as opening a translucent condiment cup and slotting a battery with 80-90% success, with only 10 minutes worth of demonstrations. Project website: tonyzhaozh.github.io/aloha

off the table. Next, one of the right fingers approaches the cup from below and pries the lid open. Each of these steps requires high precision, delicate hand-eye coordination, and rich contact. Millimeters of error would lead to task failure.

Existing systems for fine manipulation use expensive robots and high-end sensors for precise state estimation [29, 60, 32, 41]. In this work, we seek to develop a low-cost system for fine manipulation that is, in contrast, accessible and reproducible. However, low-cost hardware is inevitably less precise than high-end platforms, making the sensing and planning challenge more pronounced. One promising direction to resolve this is to incorporate learning into the system. Humans also do not have industrial-grade proprioception [71], and yet we are able to perform delicate tasks by learning from closed-loop visual feedback and actively compensating for errors. In our system, we therefore train an end-to-end policy that directly maps RGB images from commodity web cameras to the actions.

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